TOWARD A GEOMETRY OF FORMS: ORDERED AND COMBINATORIAL STRUCTURES IN GROTHENDIECK'S NOTES OF 1986

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Abstract. In June 1986 Grothendieck wrote four chapters titled *Vers une géométrie des formes*, which look for a combinatorial axiomatics of "forms", shapes in a topological sense, with no manifold structure and no triangulation. An earlier folder, *Topologie modérée*, looks for the tame subsets of a space. We state and prove the results of these notes that can be made precise.

A model of a one-dimensional form whose places make up a single segment is the same thing as a dense partial order with distinct least and greatest elements that is locally directed in both directions; the order need not be total. In such an order, the cuts made by comparable pairs determine the order up to reversal, and comparability alone does not. Cutting a space along an arbitrary subset gives a Boolean algebra of "sides" that embeds into the clopen sets of a space of local sides over the subset, with no hypothesis on the space. The saturated parts defined by a compatibility relation form a complete orthocomplemented lattice that need not be distributive, whereas the notes presume a Boolean algebra. Finally, a Boolean algebra of tame subsets admits, for every finite family, a coarsest admissible cell decomposition, and a coarsest one with connected cells. The theorems are formalised in Lean.

1. Introduction

Section 5 of the *Esquisse d'un programme* [2] calls for a "tame topology" suited to the shapes met in geometry, without the pathologies of general topology. O-minimal structures have since given one answer [3]. The University of Montpellier holds the mathematical papers Grothendieck left to Jean Malgoire, and publishes their facsimiles. Among them, five folders attempt such a theory from the combinatorial side:

- folder 156-1, *Vers une géométrie des formes (topologiques)*, chapter I, dated 5–7 June 1986: an axiomatics of one-dimensional forms by places and segments;
- folder 156-2, chapter II, 6–7 June 1986: cutting a space along a closed subset;
- folder 156-3, chapter III, 8–9 June 1986: "réseaux via découpages", forms given by their cuts;
- folder 156-4, chapter IV, 10–12 June 1986: an *analysis situs* of strata and their compatibility;
- folder 121, *Topologie modérée*: Boolean algebras of tame subsets and cell decompositions.

This paper states and proves what these notes assert and can be made precise. The statements are attributed to Grothendieck when the page states them; otherwise they are said to be *after* the page, with the change described. The proofs are ours except where we follow the page. The theorems, the lemma and the proposition are formalised in Lean 4 with Mathlib (Section 7).

2. Segments are biorders

Folder 156-1 takes a set $\mathcal{L}$ of *places*, a set of *segments*, and for each segment I its set of places $\widetilde{I} \subseteq \mathcal{L}$ and its pair of endpoints $\partial I \subseteq \widetilde{I}$. Write $I^\circ = \widetilde{I} \setminus \partial I$; a place is *regular* if it is

Date: 7 November 2026.

2020 *Mathematics Subject Classification.* 06A06; 06C15, 54E99, 57Q99, 03C64.

Key words and phrases. biorders, betweenness, partial orders, orthocomplemented lattices, tame topology, cell decompositions, Lean.

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interior to some segment. The third version of the axioms (pages 5–7) reads as follows, with $I_{x,z}$ the subsegment of I with endpoints x, z given by F1.

- F1 If $x \in \partial I$ and $z \in \widetilde{I} \backslash \{x\}$, there is a unique segment $I_{x,z}$ with $\widetilde{I}_{x,z} \subseteq \widetilde{I}$ and $\partial I_{x,z} = \{x, z\}$.
- F2 $I^\circ \neq \emptyset$; if $\partial I = \{x, y\}$ and $z \in I^\circ$, then $\widetilde{I}_{x,z} \cap \widetilde{I}_{y,z} = \{z\}$, and every segment $J \subseteq I$ with $z \in \partial J$ lies in $I_{x,z}$ or in $I_{y,z}$.
- F′3 If $x \in \partial I$ and $u, v \in I^\circ$, there is $w \in I^\circ$ with $\widetilde{I}_{x,w} \subseteq \widetilde{I}_{x,u} \cap \widetilde{I}_{x,v}$.
- F4 If z is regular, $\widetilde{I'} \cap \widetilde{I''} = \{z\}$, $\partial I' = \{x, z\}$ and $\partial I'' = \{y, z\}$, there is a unique segment I with $\widetilde{I} \supseteq \widetilde{I'} \cup \widetilde{I''}$ and $\partial I = \{x, y\}$.
- F5 A regular place has order 2: there are two segments issued from it that define different branches, and every segment issued from it defines the same branch as one of them.

Here two segments I, J issued from x (that is, $x \in \partial I \cap \partial J$) *define the same branch at* x if $I_{x,z} = J_{x,z}$ for some z interior to both. The page also states an axiom F3 on two segments issued from the same place. F′3 is a condition of a different kind, inside one segment. A note at the foot of page 6, mostly illegible, begins "Il manque" and names F'_3; the statement above is our reconstruction of it, guided by what the transitivity of "same branch" needs. Figure 1 shows F2 and F4.

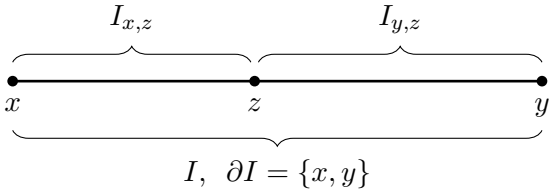


FIGURE 1. Axiom F2: for z interior to I, the subsegments $I_{x,z}$ and $I_{y,z}$ meet only at z. Axiom F4 goes the other way: two segments meeting only at a regular place z glue to a unique segment from x to y.

Page 10 calls a model a *segment model* when $\mathcal{L}$ is itself a segment I_0, $\partial I_0 = \{x_0, x_1\}$. Page 11 defines $a \leq b$ when $a = b$ or a lies on the segment from x_0 to b, and page 13 concludes that a segment model is the same thing as a *biorder*, a pair of mutually opposite orders: the other endpoint gives the opposite order. Call a partial order *locally directed downwards* if $x < y_1$ and $x < y_2$ imply $x < y \leq y_1, y_2$ for some y, and *locally directed upwards* in the dual case.

Theorem 2.1 (after folder 156-1, pp. 10–13)**.** *In a segment model satisfying F1, F2, F′3 and F4, the relation $\leq$ is a partial order with least element x_0 and greatest element $x_1 \neq x_0$; it is dense and locally directed in both directions; the segments are exactly the intervals $[a, b] = \{z \mid a \leq z \leq b\}$ with $a < b$, each carried by a single segment; and the order defined from x_1 is the opposite order.*

Conversely, let $\mathcal{L}$ be a partially ordered set with distinct least and greatest elements, dense and locally directed in both directions. The intervals $[a, b]$, $a < b$, with $\partial[a, b] = \{a, b\}$, satisfy F1–F5 and F′3; $\mathcal{L}$ is a segment, and the order read from its least element is the given one.

Sketch. Direct sense. By F1 in I_0, a segment is determined by its endpoints (using F2 when neither endpoint is x_0 or x_1). Transitivity comes from F1 inside $[x_0, c]$; antisymmetry from F2, which forbids b to lie on both sides of an interior point a. The key step is *gluing*: if a is regular and a segment Q from a to z meets $[x_0, a]$ only at a, then F4 glues them into a segment from x_0 to z, so $a \leq z$. With F2 this shows that the endpoints of every segment are comparable, and that a segment with endpoints $a < b$ is exactly the interval $[a, b]$. Density comes from $I^\circ \neq \emptyset$, the two directedness conditions from F′3 at the lower and upper endpoints.

Converse. Each axiom becomes a property of intervals: F1 and F2 by the interval structure and density, F′3 and F3 by directedness, F4 because two intervals meeting only at z must lie on opposite sides of z, and F5 by directedness and density. □

Pages 12 and 13 state both directions without proof; the clause "locally directed upwards" is written between the lines and read only in part, and the converse sits on a line with several illegible words. The direct sense uses neither F3 nor F5, and the bounds serve only to make $\mathcal{L}$ itself a segment. The pages do not say whether the order is total. It need not be.

Example 2.2. On $\mathbb{Q}^2$ let $p \prec q$ when $|q_2 - p_2| < q_1 - p_1$, the strict order of an open cone, write $p \preceq q$ for $p \prec q$ or $p = q$, and let T be the set of points p with $(0,0) \preceq p \preceq (2,0)$: the open square with diagonal from $(0,0)$ to $(2,0)$, together with these two points. It is bounded and dense (the midpoint of $p \prec q$ lies strictly between them), and locally directed both ways: if $x \prec y_1, y_2$, then $x + (\varepsilon, 0)$ works for small $\varepsilon > 0$, and dually $x - (\varepsilon, 0)$. The points $(1, \frac{1}{2})$ and $(1, -\frac{1}{2})$ of T are incomparable (Figure 2). By the converse of Theorem 2.1, the intervals of T form a segment model whose order is not total.

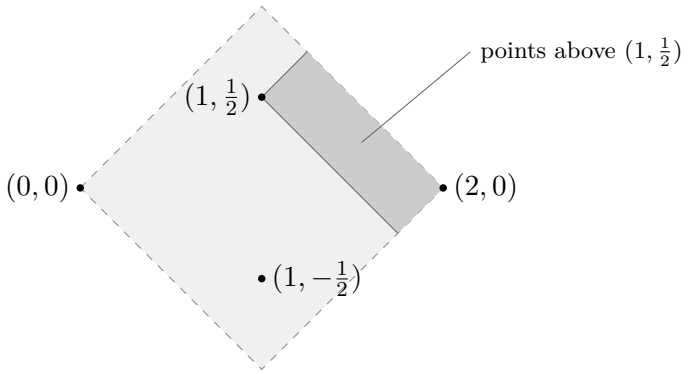


FIGURE 2. Example 2.2. The set T is the open square (light grey; its dashed edges are not in T) together with $(0,0)$ and $(2,0)$. The points of T above p form the part of T inside the open quarter-plane $p + \{|y| < x\}$; for $p = (1, \frac{1}{2})$ this is the darker open square with its corner $(2,0)$. The point $(1, -\frac{1}{2})$ lies neither in it nor below $(1, \frac{1}{2})$.

3. The order is read on the cuts

Folder 156-3 axiomatises "networks" through their cuts. Page 8 defines an *ordered stretch* (*tronçon ordonné*): a partially ordered set, dense, directed upwards and downwards, locally directed in both directions, with more than one element. For comparable $a \neq b$, let $C(\{a,b\})$ be the set of elements distinct from a, b and comparable to both; it falls into three *pieces*, the elements below both, strictly between, and above both, and the partition depends only on the pair. Write $X_{\{a,b\}}(t)$ for the piece containing t, and $\mathcal{R}(a,b,c)$ for strict betweenness ($a < b < c$ or $c < b < a$). Page 14 asks whether comparability and the partitions of the $C(\varepsilon)$ into pieces, given without labels, determine $\mathcal{R}$; page 15 answers "On gagne !".

Lemma 3.1 (after folder 156-3, p. 15)**.** *If $a < b < c$, then $X_{\{a,b\}}(c) = \mathcal{L}_{>b}$, $X_{\{b,c\}}(a) = \mathcal{L}_{<b}$ and $X_{\{c,a\}}(b) =]a,c[$. If the order is dense and p, q, r are pairwise distinct and comparable, then $\mathcal{R}(p,q,r)$ holds iff $X_{\{p,q\}}(r) \cap X_{\{q,r\}}(p) = \emptyset$.*

Proof. The three pieces are read off the definition. Three pairwise comparable elements form a chain. If q is in the middle, the two pieces are $\mathcal{L}_{>q}$ and $\mathcal{L}_{<q}$, which are disjoint. If p is in the middle, say $q < p < r$, they are $\mathcal{L}_{>p}$ and $]q,r[$, which both contain a point of $]p,r[$, non-empty by density; the other cases are similar. □

Theorem 3.2 (after folder 156-3, pp. 11–15)**.** *A bijection between two ordered stretches that respects comparability and, for every comparable pair ε, the partition of $C(\varepsilon)$ into pieces, is an order isomorphism or an anti-isomorphism. Comparability alone does not suffice: on the*

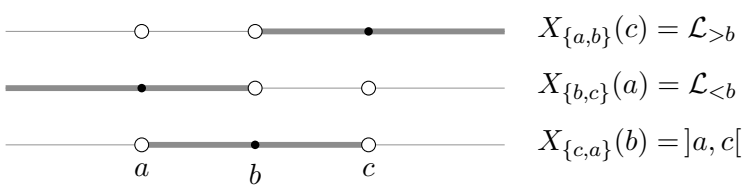


FIGURE 3. Lemma 3.1 on a chain $a < b < c$: each pair (white circles, excluded from $C(\varepsilon)$) cuts the rest into three pieces; the thick part is the piece containing the third point. The first two thick pieces are disjoint: b is between a and c.

stretch $\mathbb{Q}$, the transposition of 0 *and* 1 *respects comparability and is neither increasing nor decreasing.*

Proof. Let e be the bijection. The piece $X_{\{p,q\}}(r)$ is a class of the partition of $C(\{p,q\})$, so e maps it onto $X_{\{e(p),e(q)\}}(e(r))$; by Lemma 3.1, applied on both sides, e respects strict betweenness. Write $\overline{\mathcal{R}}(a,b,c)$ for $a \le b \le c$ or $c \le b \le a$. It is expressed by $\mathcal{R}$ and comparability: $\overline{\mathcal{R}}(a,b,c)$ holds iff $\mathcal{R}(a,b,c)$, or $b \in \{a,c\}$ and a, c are comparable. Pages 11–13 read the order on betweenness. Fix $a < b$. Then $u \le v$ iff there are x and y, with $x = a$ or $\mathcal{R}(b,a,x)$, and $y = b$ or $\mathcal{R}(a,b,y)$, such that

$$\overline{\mathcal{R}}(x,u,y), \qquad \overline{\mathcal{R}}(x,v,y), \qquad \overline{\mathcal{R}}(x,u,v).$$

Indeed, the conditions on x and y say $x \le a$ and $b \le y$. If $u \le v$, directedness gives $x \le u, a$ and $y \ge v, b$, and $x \le u \le v \le y$. Conversely $x \le a < b \le y$ gives $x < y$, so the first two conditions read $x \le u \le y$ and $x \le v \le y$; if the third reads $v \le u \le x$, then $u = x \le v$. If $e(a) < e(b)$, transporting this description along e gives $u \le v \iff e(u) \le e(v)$; if $e(b) < e(a)$, replace the order of the target by the opposite one, which has the same betweenness and comparability and is still directed both ways. If no two distinct elements are comparable, e preserves equality and there is nothing to prove. For $\mathbb{Q}$, the transposition σ has $\sigma(1) = 0 < 1 = \sigma(0)$ while $\sigma(1) = 0 < 2 = \sigma(2)$. □

Only density and directedness are used, so the theorem holds for every dense partial order in which any two elements have a common lower bound and a common upper bound; the local directedness of the definition plays no part. The second sentence of the theorem is ours. The comparability clause in the description of $\overline{\mathcal{R}}$ cannot be dropped. The margin of page 11 describes $\overline{\mathcal{R}}(a,b,c)$ as $\mathcal{R}(a,b,c)$ or $\{a,b,c\}$ of cardinality at most 2, which fails for $a = c \neq b$; the form "$\mathcal{R}(a,b,c)$ or $b \in \{a,c\}$" fails for incomparable a, c, for instance $\overline{\mathcal{R}}((0,1),(0,1),(1,0))$ is false in $\mathbb{N}^2$ with the product order. The lemma of page 13 never meets these cases, since it is applied with $x < y$. Courcelle characterises the partial orders that are determined up to reversal by their strict betweenness alone [1, Th. 3.5]; the axiomatisation of that betweenness is due to Lihová, whom he cites. His theorem covers a dense partial order with at least three elements whose comparability graph is connected, hence a dense partial order directed in both directions: for these, the comparability data of Theorem 3.2 are not needed once strict betweenness is known.

4. Sides of a subset

Folder 156-2 cuts a space along a closed subset without assuming a manifold or a triangulation. For $A \subseteq X$, $j\colon X \setminus A \hookrightarrow X$, and $\mathrm{Comp}(Z)$ the Boolean ring of clopen subsets of Z, page 14 puts

$$\mathrm{Riv}(A,X) = \varinjlim_{V \supseteq A} \mathrm{Comp}(V \setminus A),$$

the *sides* of A, the limit over neighbourhoods of A, and $\widetilde{A} = \mathrm{Spec}(j_*\mathbb{F}_2)$. Page 15 writes a canonical map $\mathrm{Riv}(A,X) \to \mathrm{Comp}(\widetilde{A})$, first "bijective if X is paracompact", then adds "injective, and" above the line. The page says neither which spectrum is meant nor which

topology it carries; we make the following choice. Encode a clopen part of $V \setminus A$ by its indicator, a function $X \to \mathbb{F}_2$ locally constant on $V \setminus A$; then $\mathrm{Riv}(A, X)$ is the ring of germs of such functions along A, and the stalk B_x of $j_*\mathbb{F}_2$ at $x \in A$ is the ring of their germs at x, taken in $X \setminus A$. Let $\widetilde{A}$ be the set of pairs (x, χ) with $x \in A$ and $\chi\colon B_x \to \mathbb{F}_2$ a ring morphism, that is, a point of the spectrum of B_x. Give it the topology generated by the sets

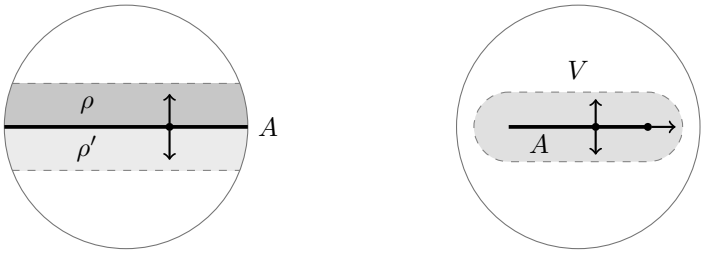
$$\{(y, \chi) \in \widetilde{A} : y \in W,\ \chi(f_y) = 1\},$$

for W open in X and f locally constant on $W \setminus A$, where f_y is the germ of f at y. The map sends a side ρ to $\{(x, \chi) : \chi(\rho_x) = 1\}$. Over a point x, the points of $\widetilde{A}$ are the ways of leaving A near x: two over each interior point of a chord of a disc, one over each end of a slit (Figure 4).

Theorem 4.1 (after folder 156-2, pp. 14–15)**.** *For every subset A of every topological space X, the image of each side is clopen in $\widetilde{A}$, the map* $\mathrm{Riv}(A, X) \to \mathrm{Comp}(\widetilde{A})$ *is a morphism of Boolean algebras, and it is injective. The ring* $\mathrm{Riv}(A, X)$ *is zero iff A is open.*

Proof. Each image is a basic open set, and the image of $1 - \rho$ is its complement, so it is clopen; products go to intersections and 1 to $\widetilde{A}$. For injectivity, let $\rho \neq \rho'$ be represented by f, f' on $V \setminus A$. Then there is $x \in A$ such that $f \neq f'$ at points of $X \setminus A$ arbitrarily close to x. Take an ultrafilter $\mathcal{U}$ finer than the trace on $\{f \neq f'\} \setminus A$ of the neighbourhood filter of x. Evaluation along $\mathcal{U}$ is a ring morphism $\chi\colon B_x \to \mathbb{F}_2$, and $\chi(\rho_x) \neq \chi(\rho'_x)$, so (x, χ) lies in exactly one of the two images. Finally $\mathrm{Riv}(A, X) = 0$ iff the filter of neighbourhoods of A meets $X \setminus A$ trivially, that is, iff some neighbourhood of A lies in A. □

Closedness of A is not used. On page 15 a *bi-side* is a pair $\{\rho, \rho'\}$ of sides with $\rho + \rho' = 1$; in a Boolean ring $\rho = \rho'$ then forces $1 = 0$. The last statement of the theorem thus corrects the page's "$A \neq \emptyset$", in the remark that the two sides of a bi-side are distinct, to "A not open". Bijectivity for paracompact X is not proved here.



chord: $\mathrm{Riv}(A, X)$ has 4 elements slit: $\mathrm{Riv}(A, X)$ has 2 elements

FIGURE 4. Theorem 4.1 for X a closed disc. Left, A is a chord from boundary to boundary: for the strip neighbourhoods V (dashed), which are cofinal, $V \setminus A$ has two components, so $\mathrm{Riv}(A, X) = \{0, 1, \rho, \rho'\}$ with $\rho' = 1 - \rho$, and $\widetilde{A}$ has two points over every point of A (arrows). Right, A is a closed segment inside the disc: for the neighbourhoods V of the shape shown, which are cofinal, $V \setminus A$ is connected, so $\mathrm{Riv}(A, X) = \{0, 1\}$, yet $\widetilde{A}$ has two points over each interior point of A and one over each end.

5. Saturated parts are not a Boolean algebra

Folder 156-4 starts from a stratified space and abstracts a partially ordered set $(\mathcal{M}, \leq)$ of *multistrata* with a symmetric relation R, *compatibility*, reflexive and inherited downwards. On pages 65–68, two multistrata are *disjoint* if they are compatible and have no common lower bound; the *cosupport* Φ' of $\Phi \subseteq \mathcal{M}$ is the set of multistrata disjoint from every element of Φ, the *support* is Φ'', and Φ is *saturated* if $\Phi = \Phi''$. Page 68 presumes that the saturated parts form a Boolean algebra, even an "ultra-Stonean" one.

Proposition 5.1 (after folder 156-4, pp. 66–67)**.** *For any partially ordered set $(\mathcal{M}, \leq)$ with a symmetric relation R, intersections of saturated parts are saturated, every cosupport is saturated, and on saturated parts the cosupport is an inclusion-reversing involution with $S \cap S' = \emptyset$ and $(S \cup S')'' = \mathcal{M}$. The saturated parts form a complete orthocomplemented lattice.*

This is the classical lattice of closed sets of a polarity. For orthogonality in a finite-dimensional Euclidean space the closed sets are the linear subspaces, and they do not form a Boolean algebra. The next theorem shows that the axioms of the folder allow the same behaviour.

Theorem 5.2 (counterexample to folder 156-4, p. 68)**.** *Let $\mathcal{M}$ have four minimal elements a, b, c, d and three elements v_{ab}, v_{bc}, v_{cd}, each above the pair it indexes, and call two elements compatible when they lie under a common element. Then $\leq$ is an order, R is reflexive, symmetric and inherited downwards, and the saturated parts are exactly*

$$\emptyset, \quad \{b\}, \quad \{c\}, \quad \{a, c\}, \quad \{b, d\}, \quad \mathcal{M}.$$

They form a hexagon, which is not distributive: $\{b, d\} \wedge (\{b\} \vee \{c\}) = \{b, d\}$, while $(\{b, d\} \wedge \{b\}) \vee (\{b, d\} \wedge \{c\}) = \{b\}$.

Proof. Two elements are compatible when they lie under a common element, and disjoint when moreover nothing lies under both. The disjoint pairs are therefore $\{a, b\}$, $\{b, c\}$, $\{c, d\}$: each v has a common lower bound with each element it is compatible with. So a part containing some v has empty cosupport, $\emptyset' = \mathcal{M}$, $\mathcal{M}' = \emptyset$, and the cosupport of a nonempty part of $\{a, b, c, d\}$ is the set of common neighbours of its elements in the path $a - b - c - d$. Thus $\{b\}' = \{a, c\}$, $\{a, c\}' = \{b\}$, $\{c\}' = \{b, d\}$, $\{b, d\}' = \{c\}$; $\{a\}'' = \{b\}' = \{a, c\}$ and $\{d\}'' = \{c\}' = \{b, d\}$; a part of $\{a, b, c, d\}$ with two or more elements other than $\{a, c\}$ and $\{b, d\}$ contains two adjacent vertices or $\{a, d\}$, has empty cosupport and support $\mathcal{M}$. This gives the list; Lean checks it on all 128 parts. The join of $\{b\}$ and $\{c\}$ is $\{b, c\}'' = \mathcal{M}$. □

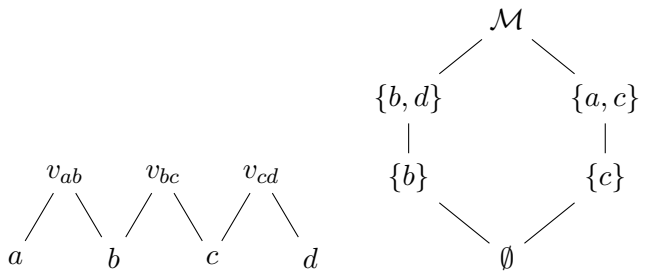


FIGURE 5. Theorem 5.2: the seven-element model and the hexagon lattice of its saturated parts, which is orthocomplemented ($\{b\} \leftrightarrow \{a, c\}$, $\{c\} \leftrightarrow \{b, d\}$) but not distributive.

Whether the later disjointness axioms of the folder (C 5, page 69, and the "packet of disjointness axioms" of page 85) restore distributivity is not decided here. When R is always true, the saturated parts do form a complete Boolean algebra (a classical fact, not formalised here). Give $\mathcal{M}$ the topology whose open sets are the down-sets. Two elements are disjoint iff their down-sets do not meet, so $\Phi' = \mathcal{M} \setminus \overline{\downarrow\Phi}$, and $\Phi'' = \operatorname{int} \overline{\downarrow\Phi}$: the saturated parts are the regular open sets.

6. Coarsest cell decompositions in a tame Boolean algebra

Folder 121, *Topologie modérée*, looks for a notion of tame subset of a topological space X. Pages 3–5 take a Boolean algebra $\mathcal{F}$ of subsets of X and the conditions: (c bis) $A \in \mathcal{F} \Rightarrow \overline{A} \in \mathcal{F}$; (d) every $A \in \mathcal{F}$ contains a dense open subset of its closure, open relative to the closure;

(e) every decreasing sequence of closed members of $\mathcal{F}$, each nowhere dense in the previous one, reaches $\emptyset$. Page 8 shows that the frontier of a member is nowhere dense. Page 10 calls a *cell decomposition* a finite partition of X such that the closure of each cell is a union of cells and distinct cells have distinct closures; it is *$\mathcal{F}$-admissible* if its cells are in $\mathcal{F}$. A decomposition is *good* for a family $\mathcal{L}$ if it is $\mathcal{F}$-admissible and every member of $\mathcal{L}$ is a union of cells; it is *coarser* than another if every cell of the other lies in one of its cells. The same definitions apply with a closed member Z in place of X. In (d) the page writes the interior in X; read literally this would exclude points, which the cell decompositions need, so we take it relative to the closure, as above.

Theorem 6.1 (Grothendieck, folder 121, pp. 14 and 16)**.** *Let $\mathcal{F}$ satisfy (c bis), (d) and (e). Every finite family $\mathcal{L} \subseteq \mathcal{F}$ has an $\mathcal{F}$-admissible cell decomposition in which each member of $\mathcal{L}$ is a union of cells, and among these a coarsest one. If moreover (f) for closed $B, C \in \mathcal{F}$ the connected components of $B \setminus C$ are finite in number and belong to $\mathcal{F}$, the same holds for decompositions with connected cells.*

Proof. We prove the statement with X replaced by any closed member Z, by induction along (e), following page 14. Let $X' \subseteq Z$ be the union of the frontiers $\overline{Y \cap Z} \cap \overline{Z \setminus Y}$, $Y \in \mathcal{L}$. It is a closed member, nowhere dense in Z by page 8. On $Z \setminus X'$, the points with the same pattern of membership in the members of $\mathcal{L}$ form the *open cells*; a point of $Z \setminus X'$ in the closure of an open cell has its pattern, so the closure of an open cell U meets no other open cell and $U = \overline{U} \setminus X'$. By induction, X' has a coarsest good decomposition for the family $\{Y \cap X'\}_{Y \in \mathcal{L}} \cup \{\overline{U} \cap X'\}_U$; the second part of this family gives the frontier condition between open cells and cells of X'. The open cells together with that decomposition of X' are a good decomposition of Z.

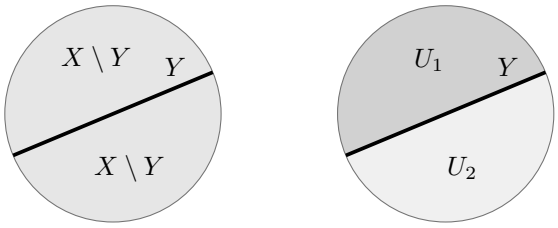
Page 14 asserts without argument that it is the coarsest. Our argument: in a decomposition in which Y is a union of cells, a cell that meets the frontier of Y is contained in it. So every good decomposition of Z splits into cells inside X', which form a good decomposition of X' for the enlarged family, and cells inside $Z \setminus X'$, each contained in an open cell. Both parts are therefore finer than ours.

For connected cells (page 16), replace each open cell U by its connected components. Since $U = \overline{U} \setminus X'$ with $\overline{U}$ and X' closed members, (f) makes them finitely many members of $\mathcal{F}$. A component V of U satisfies $\overline{V} \cap U \subseteq V$, which gives the frontier condition, and a connected cell of a competing decomposition that lies in U lies in one component. □

For example, let X be a closed disc in $\mathbb{R}^2$ and $\mathcal{F}$ the semialgebraic subsets of X. Conditions (c bis) and (f) are standard: the closure of a semialgebraic set is semialgebraic, and a semialgebraic set has finitely many connected components, each semialgebraic [3, Ch. 2 and 3]. For (d), let $B = \overline{A} \setminus A$ and suppose $\overline{B}$ contains a nonempty semialgebraic set W open in $\overline{A}$. Then $A \cap W$ is dense in W and $B \cap W$ lies in the frontier of $A \cap W$, so $\dim W \leq \dim(B \cap W) < \dim(A \cap W) \leq \dim W$, by the inequality $\dim(\overline{S} \setminus S) < \dim S$ [3, Ch. 4]; this is absurd. For (e), if a closed semialgebraic set Z' is nowhere dense in a closed semialgebraic set Z, then $Z \setminus Z'$ is dense in Z, so Z' lies in the frontier of $Z \setminus Z'$ and $\dim Z' < \dim Z$; the dimension drops at each step of the chain. Figure 6 shows the two decompositions of the theorem for a chord of the disc.

7. Verification

The results above are formalised in Lean 4 with Mathlib, using only Lean's standard axioms. The files are in the directory `lean/Grothendieck` of the repository Commutator-IO/grothendieck-project; the table lists the main declarations.



page 14: $\{X \setminus Y,\ Y\}$ page 16: $\{U_1,\ U_2,\ Y\}$

FIGURE 6. Theorem 6.1 for X a closed disc, $\mathcal{F}$ its semialgebraic subsets and $\mathcal{L} = \{Y\}$, Y a closed chord from boundary to boundary. The frontier of Y is Y itself. Left, the coarsest admissible decomposition: two cells, $X \setminus Y$ (not connected) and Y. Right, the coarsest one with connected cells: $X \setminus Y$ splits into its components U_1, U_2, with $\overline{U_i} = U_i \cup Y$.

Result	File	Declarations
Theorem 2.1	`Folder156_1`	`SegModel.biorder`, `SegModel.le_swap`, `intervals_axioms`, `intervals_segModel`, `intervals_le_iff`
Example 2.2	`Folder156_1`	`Diamond.not_total`, `Diamond.example_segModel`
Lemma 3.1	`Folder156_3`	`piece_above`, `piece_below`, `piece_between`, `middle_iff`
Theorem 3.2	`Folder156_3`	`btw_transport`, `le_iff_readLe`, `order_eq_or_dual`, `order_of_cuts`, `cmp_alone_insufficient`, `btwLe_iff`
Theorem 4.1	`Folder156_2`	`isOpen_toSpecSet`, `toSpecSet_compl`, `toComp_mul`, `toComp_one`, `toComp_injective`, `riv_one_eq_zero_iff`
Proposition 5.1	`Folder156_4`	`isSaturated_iInter`, `isSaturated_cosupp`, `cosupp_cosupp`, `inter_cosupp_eq_empty`, `supp_union_cosupp`
Theorem 5.2	`Folder156_4`	`hexagon_hyp`, `hexagon_saturated`, `hexagon_not_distrib`
Theorem 6.1	`Folder121`	`Tame.rare_bd`, `subset_bd`, `exists_coarsest`, `page14`, `exists_coarsest_conn`, `page16`

Not formalised: the Boolean case at the end of Section 5 and the semialgebraic example of Section 6. Not proved here at all: the corollaries of F1, F2, F4 outside segment models, restriction and orientation (folder 156-1, pp. 8–17); the bijectivity in Theorem 4.1 for paracompact spaces and the cut space of pages 16–17; the axioms of networks by cuts (folder 156-3, pp. 1–5 and 16–39); the later disjointness axioms of folder 156-4; the regular decompositions of folder 121, page 17.

Statement on the use of artificial intelligence

The transcriptions of the folders, the modernised readings, the proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the statements, the proofs and the references, and takes full responsibility for the content. The formal proofs are checked mechanically by Lean.

Acknowledgements

The facsimiles are published by the University of Montpellier, grothendieck.umontpellier.fr.

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